

TASK 05-50-06-200-801-A

EFFECTIVITY: ALL

1. Maximum Operating Speed Envelope Exceedance - Inspection/Check

A. General

- (1) This section gives the necessary actions after an OVSPD event.
- (2) The exceedance of the operating speed/Mach envelope will trigger the OVSPD aural warning.
- (3) Do an OVSPD inspection when the crew reports trigger of the overspeed aural warning or when the flight data analysis shows that the operating speed envelope was exceeded.
- (4) The procedure considers and VMO exceedance, defined as follows:
 - VMO is the maximum operating speed
 - MMO is the maximum operating Mach

NOTE: An excess of any of these two parameters is considered an OVSPD event.

- (5) The TCRF may be used to check if there are any triggered event.
- (6) The operating speed envelope is presented in the AFM - Section 2 - Limitations, is reproduced in [Figure 601](#), together with the thresholds and exceedance zones defined on this procedure.
- (7) Based on the speed envelope, two threshold values are set to classify the severity of the OVSPD event. The severity of the OVSPD will classify the event in one of the situations below:
 - (a) Below Low-Threshold value: The low-threshold value defines an OVSPD limit for which no inspection of the aircraft is necessary, if two other requirements (OVSPD duration and number of occurrences per flight) are observed, see [Figure 601](#).
 - (b) Between Low and High-Threshold values: This region defines a situation clearly out of the aircraft envelope. But the operator can still restore the aircraft airworthiness, without Embraer's involvement, if no discrepancies are found during the applicable inspections, see [Figure 601](#).
 - (c) Above High-Threshold value: The high-threshold value defines an excess to the aircraft operating speed in a high degree of severity. In this condition, the operator still has the necessary information to restore the aircraft airworthiness, but Embraer must be informed and receive data, see [Figure 601](#).
- (8) The purpose of this section is to give the operator all information to analyze the flight data, determine the necessary inspections and restore the aircraft airworthiness. Embraer must be involved only if at least one of the situations below occurs:
 - (a) The OVSPD event occurs above the high-threshold value. In this situation, the data must be sent to Embraer, but the operator is still responsible to ensure, based on the required inspections, that the aircraft can return to service, or;

(b) During any phased inspection, a discrepancy is found. In this situation, the operator sends to Embraer the data and a report of the discrepancies found. An Embraer's determination is necessary to return the aircraft to service.

(9) These maintenance technicians are necessary to do this task:

- 1 - Electrical/Avionics Systems
- 1 - Mechanical Systems
- 1 - Airframe (Structure)

B. References

REFERENCE	DESIGNATION
AMM MPP 05-50-05/600	HIGH LOAD FACTOR - INSPECTION/CHECK
AMM TASK 07-10-01-500-801-A/200	Complete Aircraft Jacking - Maintenance Practices
AMM TASK 12-11-01-650-802-A/300	Fuel Tank - Pressure Defueling
AMM TASK 20-00-00-910-801-A/200	Aircraft Safety Procedures Before Maintenance - Standard Procedures
AMM TASK 27-10-00-710-801-A/500	Aileron Control System - Operational Test
AMM TASK 27-20-00-710-801-A/500	Rudder Control System - Operational Test
AMM TASK 27-30-00-710-801-A/500	Elevator Control System - Operational Test
AMM TASK 27-40-00-710-801-A/500	Horizontal Stabilizer System - Operational Test
AMM TASK 27-50-00-710-801-A/500	Flap Control System - Operational Test
AMM TASK 27-60-00-710-801-A/500	Spoiler System - Operational Test
AMM TASK 27-80-00-710-801-A/500	Slat Control System - Operational Test
AMM TASK 32-12-00-820-801-A/200	Main-Landing-Gear Doors - Adjustment
AMM TASK 32-22-01-820-801-A/200	Nose-Landing-Gear Forward-Door Rods Linkage - Adjustment
AMM TASK 32-22-03-820-801-A/200	Nose-Landing-Gear Aft-Door Rods Linkage - Adjustment
AMM TASK 51-50-00-820-801-A/200	Aircraft Alignment Check

C. Overspeed Inspection

SUBTASK 210-002-A

- (1) The necessary actions after an OVSPD event are given in the flowchart shown in [Figure 602](#).
- (2) When the crew reports the OVSPD aural warning trigger, the aircraft can return to service without any other action if all the conditions that follow are correct:
 - (a) The speed stayed below 330 KIAS, and the Mach stayed below M 0.83, for all flight ratings;
 - (b) The OVSPD event did not last for more than 10 s.
 - (c) There was only one OVSPD event in the same flight.

- (3) If the crew cannot assure that all the three conditions above were respected, it is necessary to do the fight data download and data must be analyzed.
- (4) If it is not possible to do the fight data analysis immediately, do Phase-I inspection. If no defect is detected, a deferrable period of 10 FC is permitted.

NOTE: The TCRF may be used to check if there are any triggered event.

- (5) In the deferrable period, the fight data must be analyzed and compared with the threshold values. This will give you the necessary actions.

NOTE: At the end of the deferrable period, if the fight data analysis is not available, do the Phase-II inspection.

- (6) When the detection source for a maximum operating speed exceedance is flight data analysis, not a crew report, you must do the required inspections as given in the [Figure 602](#) flow chart.

NOTE: After the exceedance is reported through the flight data analysis, a deferrable period of 10 FC is granted and during this period, do the Phase I inspection.

- (7) For the event assessment, the parameters below must be selected from the fight data source:

- Time (GMT)
- Indicated Airspeed (Airspeed-Pil and Airspeed-CoP)
- Indicated Mach (Mach)
- Normal acceleration (NormAccel)
- Pressure Altitude

NOTE: The data must be selected from the fight data source at a rate of 8 sps (samples per second).

D. Maximum Operating Speed or Maximum Mach Operation Exceedance Detection

SUBTASK 210-003-A

- (1) In the fight data downloaded, identify the region where the OVSPD occurred.
- (2) Select the highest measured sample for the M (Mach) and KIAS. Compare it with the threshold values given in the [Table 601](#):

Table 601 - THRESHOLD VALUES

PARAMETER	LOW THRESHOLD (T _L)	HIGH THRESHOLD (T _H)
KIAS	330 Knots	350 Knots
M	M 0.83	M 0.85

- (3) Refer to [Table 602](#) for the necessary actions, based on the value of the most critical sample.

Table 602 - MOST-CRITICAL SAMPLE AND REQUIRED ACTIONS

MOST-CRITICAL SAMPLE POSITION	REQUIRED ACTIONS
Below T_{low} (≤ 330 KIAS or M 0.83)	No action is required and the aircraft can return to service if DVDR data shows that both conditions below were respected: - There was only one OVSPD event in the same flight: - The OVSPD event did not last for more than 10 s. NOTE: If one of these conditions was not respected, consider it as an event between T_{low} and T_{high}.
Between T_{low} and T_{high} (330 to 350 KIAS or M 0.83 to 0.85)	1. Do the Phase-II inspection during the deferrable period. 2. If no discrepancy is found during Phase-II inspection, return the aircraft to service. If you find any, do the Phase-III inspection and contact Embraer. 3. Send the DVDR data and the inspection results to Embraer if any discrepancy is found during the inspection Phases I, II or III.
Above T_{high} (≥ 350 KIAS or M 0.85)	1. Do the Phase-I inspection before the next flight. 2. If no discrepancy is found during Phase-I inspection, dispatch the aircraft for 10 FC. If you find any, contact Embraer. 3. Do the Phase-II and Phase-III inspections during the deferrable period. 4. If no discrepancy is found during Phase-II and Phase-III inspections, return the aircraft to service. If you find any, contact Embraer.

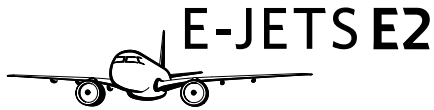
- (4) Do a check, for the interval containing the speed off-envelope points, for a possible Normal Acceleration threshold exceedance. The threshold for this condition is 2 g and if an exceedance occurs, do AMM MPP 05-50-05/600.

NOTE: The check of a possible excess to the Normal Acceleration parameter, during an OVSPD incident, is necessary as the threshold for this parameter decreases from 2.5 g to 2 g.

E. Job set-Up

SUBTASK 940-002-A

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WARNING: MAKE SURE THAT THE AIRCRAFT IS IN A SAFE CONDITION BEFORE YOU DO THE MAINTENANCE PROCEDURES. THIS IS TO PREVENT INJURY TO PERSONS AND/OR DAMAGE TO THE EQUIPMENT.

- (1) Do the procedure to make the aircraft safe for maintenance (AMM TASK 20-00-00-910-801-A/200).

F. Phase-I Inspection

SUBTASK 210-004-A

- (1) Do an external visual inspection of the entire aircraft for missing components, such as doors, access panels and fairings.
- (2) Do an inspection of the aircraft doors and access panels for distortion, broken or missing latches, and missing fasteners.
- (3) Do a visual inspection of the aileron assembly for general condition.
- (4) Do a visual inspection of the rudder assembly for general condition.
- (5) Do a visual inspection of the elevator assembly for general condition.
- (6) Do a visual inspection of the inboard and outboard flaps assemblies for general condition.
- (7) Do a visual inspection of the slat assembly for general condition.
- (8) Do a visual inspection of the spoilers assemblies for general condition.
- (9) Make sure that the aileron, rudder, elevator, flaps, slats and spoilers move freely.

G. Phase-II Inspection

SUBTASK 210-005-A

- (1) Fuselage Inspection
 - (a) Do an external visual inspection of the fuselage skin (bottom, side and crown) for distortion, buckling and pulled or missing rivets.
 - (b) Carefully examine the areas comprised between frames 43 and 58 and between frames 90 and 100.

NOTE: If you find any discrepancy during the external inspection, examine the internal structure for damage. In this case, look for damage on the skin, ribs, frames and attachments.

- (c) Examine the internal structure of the rear fuselage, between frames 90 and 100. Look for distortion, buckling and pulled or missing rivets, with particular attention to:
 - aft pressure bulkhead;
 - machined frames;
 - pitch trim actuator and its frames attachments;
 - hinges and fittings.

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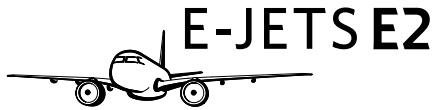
- (d) Examine the area adjacent to the fuselage bulkheads in the attachment region of wing spars I, II and III, and empennage. If you find any external damage, examine all the internal structure in the damaged areas.
 - (e) Examine the seal fairing of wing-to-fuselage fairing side panel.
 - (f) Inspect the windshield wiper for general condition.
- (2) Wing Inspection
- (a) Externally examine the upper and lower wing surfaces. Look for general misalignment, missing or pulled fasteners. If you find any discrepancy, examine the internal structure for damage.
 - (b) Examine the external surface of the trailing edge for buckles. If you find any discrepancy, examine the internal structure for damage.
 - (c) Examine the winglet surfaces for distortion and missing or pulled fasteners.
 - (d) Examine the external surface of the wing-to-fuselage fairing. Look for signs of distortion and missing or pulled fasteners.
 - (e) Do a inspection of the wing stub spar III.
 - (f) Inspect the MLG bay for signs of fuel leaks.
- (3) Empennage Inspection
- (a) Do an inspection of the horizontal stabilizer lower and upper forward and aft splice-plates.
 - (b) Do an inspection of the horizontal stabilizer upper and lower skin.
 - (c) Do an inspection of the vertical stabilizer-to-rear fuselage attachment.
 - (d) Deflect the rudder and elevator and examine the external spars. Look for distortion, buckling, missing or pulled fasteners. If you find any discrepancy, examine the internal structure for damage.
 - (e) Examine the dorsal fin external surface for signs of buckling.
 - (f) Check the horizontal and vertical stabilizers tip fairings for cracks and missing fasteners.
 - (g) For the elevator and rudder, check the hinges for integrity.
 - (h) Examine the aerodynamic seals for integrity.
 - (i) Examine the access panels for general condition and missing fasteners.
- (4) Flight Controls Inspection
- (a) Do an inspection of the elevator upper and lower skins.
 - (b) Do an inspection of the elevator hinges, fittings and actuator.
 - (c) Do an inspection of the rudder skin.

- (d) Do an inspection of the rudder hinges, fittings and actuators.
 - (e) Do an inspection of the aileron skin.
 - (f) Do an inspection of the aileron hinges, fittings and actuator.
 - (g) Do a check of all the surfaces for correctly rigging and alignment.
 - (h) Do an inspection of the rudder leading edge for signs of chafing with the vertical empennage fairing.
 - (i) Examine the aerodynamic seals of all flight-control surfaces for integrity.
 - (j) Examine the access panels for general condition and missing fasteners.
- (5) MLG and NLG Doors
- (a) Do a check of the doors for general misalignment (and);
 - (b) Do a check of the door mechanism for bowed or distorted rods;
 - (c) Do a check of the rigging of the assembly, as follows:
 - For MLG doors (AMM TASK 32-12-00-820-801-A/200);
 - For NLG. NLG FWD doors (AMM TASK 32-22-01-820-801-A/200), and NLG AFT doors (AMM TASK 32-22-03-820-801-A/200).
- (6) Pylons and Nacelles
- (a) Do an inspection of the pylons access panels for buckling, cracks, missing or pulled fasteners.
 - (b) Do a check of the aerodynamic seals for general condition.
 - (c) Do an inspection of the nacelles for signs of fuel and other fluid leaks.

H. Phase-III Inspection

SUBTASK 210-006-A

- (1) Do the operational checks of the:
 - Aileron (AMM TASK 27-10-00-710-801-A/500);
 - Inboard and outboard flaps (AMM TASK 27-50-00-710-801-A/500);
 - Slats (AMM TASK 27-80-00-710-801-A/500);
 - Spoilers (AMM TASK 27-60-00-710-801-A/500);
 - Rudder (AMM TASK 27-20-00-710-801-A/500);
 - Horizontal stabilizer (AMM TASK 27-40-00-710-801-A/500);
 - Elevator (AMM TASK 27-30-00-710-801-A/500).
- (2) Defuel the aircraft (AMM TASK 12-11-01-650-802-A/300).



- (3) Jack the aircraft (AMM TASK 07-10-01-500-801-A/200).
- (4) Do a check of the aircraft alignment (AMM TASK 51-50-00-820-801-A/200).

I. Job Close-Up

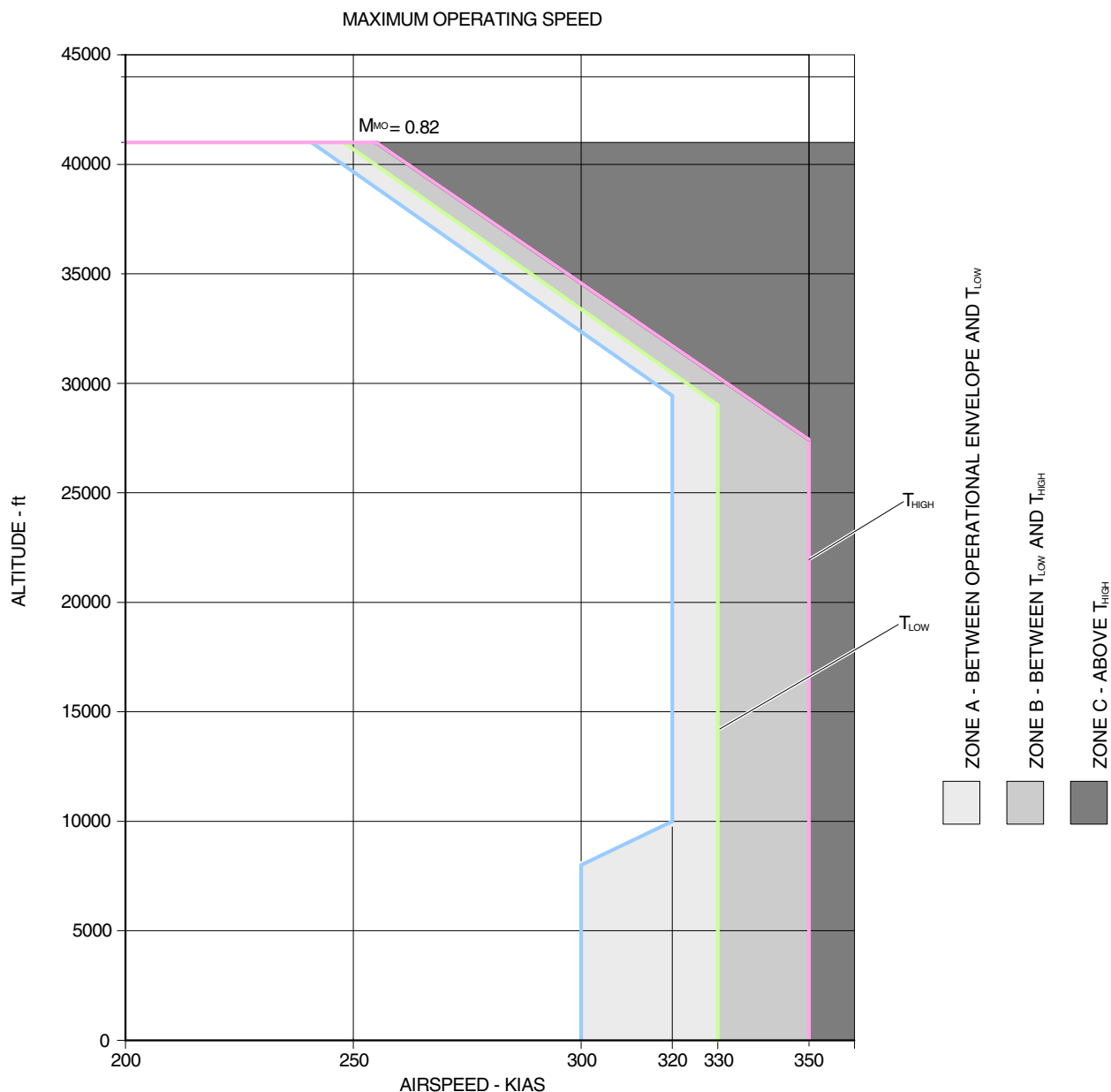
SUBTASK 940-003-A

- (1) Put the aircraft back to its initial condition (AMM TASK 20-00-00-910-801-A/200).

EFFECTIVITY: ALL

Operational Speed Envelope

Figure 601



	ALTITUDE (ft)	MAXIMUM OPERATING 320 KIAS OR M0.82	LOW THRESHOLD 330 KIAS OR M0.83	HIGH THRESHOLD 350 KIAS OR M0.85
TRANSITION KIAS / MACH	28888	320.0	324.2	332.9
	28055	320.0	330.0	338.8
	26490	320.0	330.0	350.0
TRANSITION 300 / 320 KIAS	10000	320.0	330.0	350.0
	8000	300.0	330.0	350.0

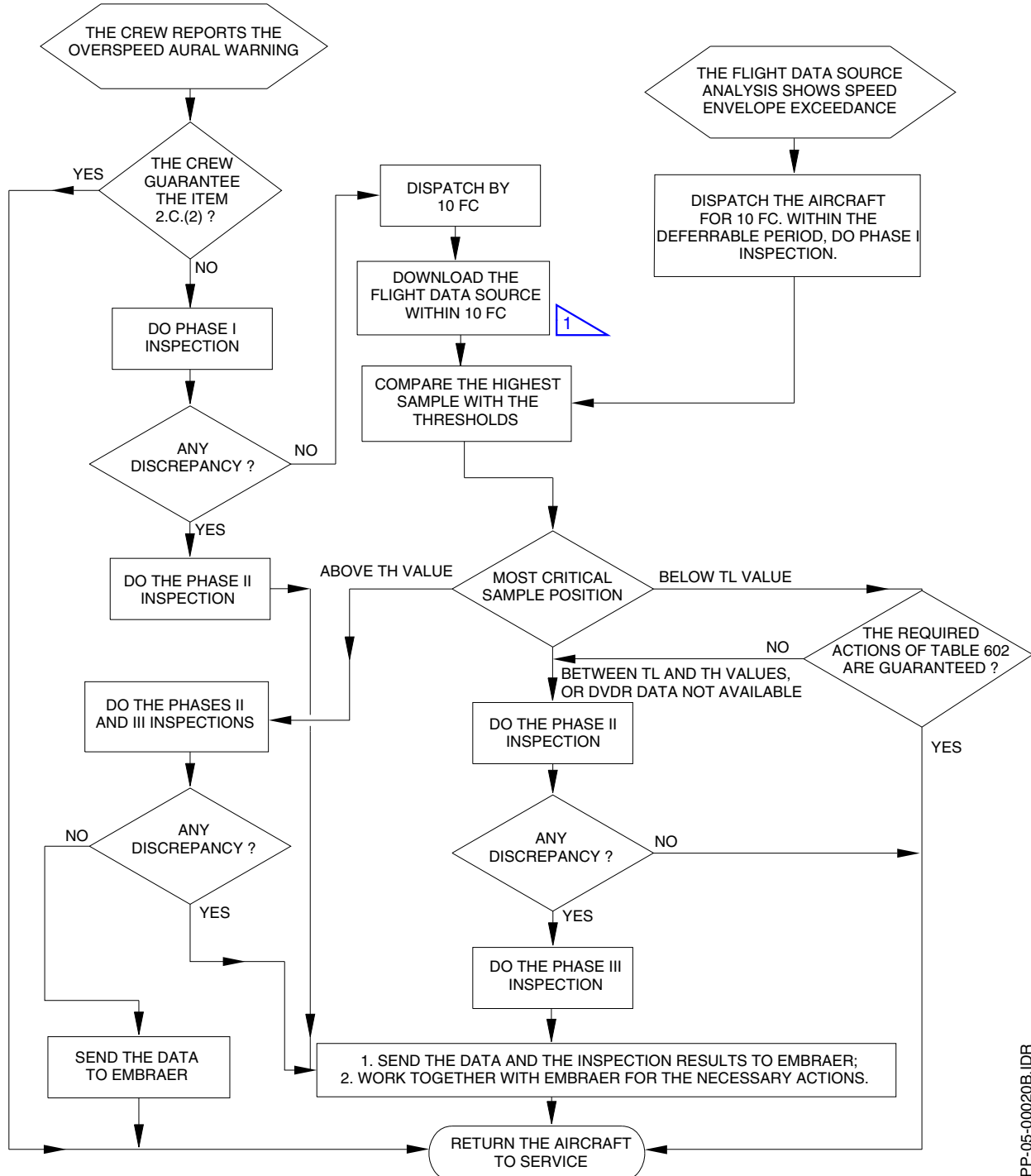
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EFFECTIVITY: ALL

Overspeed Detection and Inspection Flowchart

Figure 602



1 DO THE FLIGHT DATA DOWNLOAD IN A MAXIMUM OF 20 FH, OTHERWISE YOU CAN MISS THE DATA BECAUSE OF THE RECORDING TIME OF THE UNIT.

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